

Healthcare Operations Dashboard Report

Project Title

Healthcare Incident & Operations Analytics Dashboard

1. Project Overview

This project focuses on analyzing healthcare operational incidents using SQL and Power BI to identify operational bottlenecks, analyst performance trends, SLA efficiency, client workload distribution, and recurring system failures.

The dashboard was designed to simulate a real-world healthcare operations monitoring environment where incident management, service delivery, and operational efficiency are critical.

The project includes:

- SQL-based data analysis
 - KPI development
 - SLA analytics
 - Root Cause Analysis (RCA)
 - Analyst performance tracking
 - Client operations monitoring
 - Interactive Power BI dashboards
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2. Business Problem

Healthcare organizations depend heavily on operational systems such as:

- Claims processing
- Eligibility verification
- ETL pipelines
- API integrations
- Payment posting systems
- Database synchronization

Operational incidents in these systems can directly affect:

- Patient services
- Insurance workflows
- Financial operations

- Reporting accuracy
- Service-level agreements (SLAs)

The objective of this project was to build a centralized analytics solution to:

- Monitor incident trends
- Track SLA performance
- Identify recurring root causes
- Evaluate analyst efficiency
- Understand client operational load
- Improve operational decision-making

3. Tools & Technologies Used

Tool	Purpose
SQL	Data extraction and analysis
MySQL	Database management
Power BI	Dashboarding and visualization
DAX	KPI and measure creation
Excel/CSV	Dataset preparation

4. Dataset Overview

The dataset contains operational healthcare incident records.

Main Tables Used

Incidents

Contains:

- Incident ID
- Incident Category
- Root Cause
- Priority
- Status
- Resolution Time
- Analyst ID
- Client ID

- Created Date

Analysts

Contains analyst information:

- Analyst ID
- Analyst Name

Clients

Contains client details:

- Client ID
 - Client Name
 - Region
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5. SQL Analysis Performed

The project included multiple levels of SQL analysis:

Basic Analysis

- Open incident tracking
- High-priority incident identification
- Incident count by status
- Incident count by priority
- Average resolution time

Intermediate Analysis

- Top root causes
- Top analysts
- Top clients
- SLA classification
- Incident category analysis
- Above-average analyst performance

Advanced SQL Concepts Used

- Common Table Expressions (CTEs)
- CASE statements
- Aggregate functions
- Subqueries
- GROUP BY and HAVING

- Conditional aggregation
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6. Dashboard Pages

Page 1 — Operations Overview Dashboard

Objective

Provide a high-level operational snapshot.

KPIs Included

- Total Incidents
- Open Incidents
- Resolved Incidents
- Average Resolution Time
- Critical Incidents
- Active Analysts

Key Visuals

- Incident Count by Status
- Incident Count by Priority
- Top Root Causes
- Top Analysts by Incident Handling

Insights

- More than half of incidents were resolved successfully.
 - Medium and High priority incidents contributed the largest operational workload.
 - API Failure and Database Lock were the most recurring root causes.
 - A small group of analysts handled the majority of incidents.
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Page 2 — SLA & Resolution Analytics

Objective

Analyze operational efficiency and SLA performance.

KPIs Included

- Slow SLA Incidents
- Fast SLA Percentage
- Most Common Root Cause
- Average Resolution Time

Key Visuals

- SLA Category Distribution
- Avg Resolution Time by Analyst
- Incident Trend Over Months
- Resolved vs Open vs In Progress by Analyst
- Root Cause by Incident Category Matrix

Insights

- Approximately 60% of incidents were resolved within the Fast SLA range.
 - Some analysts showed significantly higher average resolution times, indicating potential workload imbalance.
 - Incident spikes were observed during specific months, suggesting seasonal or operational pressure.
 - API Failures and Database Locks consistently appeared across multiple incident categories.
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Page 3 — Client & Operations Analysis

Objective

Evaluate client operational impact and analyst efficiency.

KPIs Included

- Active Analysts
- Top Analyst
- Top Client
- Avg Incidents per Analyst

Key Visuals

- Analyst Efficiency Scatter Plot
- Jobs Causing Most Incidents
- Incident Volume by Region
- Top Clients by Incident Volume
- Client vs Priority Analysis

Insights

- NovaCare Systems generated the highest number of operational incidents.
 - North America contributed the highest regional incident volume.
 - ETL pipelines, API pulls, and payment reconciliation jobs were major operational risk areas.
 - Analyst workload distribution was uneven, with a few analysts handling significantly more incidents.
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7. Key Business Insights

Operational Insights

- Incident concentration was highest in Medium and High priority levels.
- Certain root causes repeatedly impacted multiple operational areas.
- Operational spikes indicate possible system capacity or workflow issues.

SLA Insights

- Fast SLA compliance remained strong overall.
- Slow SLA incidents still represented a critical operational risk.
- Some analysts may require workload redistribution to improve response times.

Client Insights

- A few clients contributed disproportionately high operational workload.
- High-volume clients may require dedicated operational support.
- Regional concentration suggests infrastructure or client usage imbalance.

Analyst Insights

- Top-performing analysts resolved significantly more incidents.
 - Analyst efficiency varied across teams.
 - Resolution time analysis can support workforce optimization.
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8. Recommendations

Based on the analysis, the following operational recommendations are suggested:

1. Reduce Recurring Root Causes

Focus on:

- API stability improvements
- Database optimization
- ETL monitoring enhancements

2. Improve SLA Performance

- Rebalance analyst workload
- Prioritize critical incidents faster
- Introduce automated escalation alerts

3. Client-Specific Monitoring

High-volume clients should have:

- Dedicated support tracking
- Priority monitoring
- Preventive maintenance reviews

4. Analyst Performance Optimization

- Identify overloaded analysts
 - Improve ticket assignment logic
 - Introduce analyst performance monitoring KPIs
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9. Project Outcome

This project successfully demonstrated how SQL and Power BI can be combined to create a complete healthcare operations analytics solution.

The dashboard enables:

- Operational monitoring
 - SLA tracking
 - Root cause identification
 - Analyst performance analysis
 - Client workload analysis
 - Executive-level decision support
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10. Skills Demonstrated

SQL Skills

- Joins
- Aggregations
- Subqueries
- CASE statements
- CTEs
- Analytical queries

Power BI Skills

- KPI card creation
- Interactive slicers
- Dashboard design
- DAX measures
- Drill-down analysis
- Data storytelling

Business Skills

- Operational analytics
 - SLA monitoring
 - Root cause analysis
 - Healthcare operations understanding
 - Performance analytics
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11. Conclusion

The Healthcare Operations Dashboard project provides a strong example of operational intelligence and business analytics using SQL and Power BI.

The solution helps stakeholders:

- monitor operational health,
- identify recurring issues,
- improve SLA compliance,
- optimize analyst performance,
- and support data-driven operational decisions.

This project demonstrates practical business intelligence capabilities applicable to healthcare analytics, operations analytics, and enterprise support environments.